Smarter control of home solar, batteries and EVs

What a forecast-driven controller is worth per home in Portugal and Spain: a year-long simulation on 2025 market data and 2026 tariffs.

€145–190
saved per year per solar + battery + EV home on time-of-use or hourly tariffs, paid market prices for exports, against default settings (€77 on a flat tariff)

€45–100
saved per year per solar + battery home without an EV, same conditions

40–70%
of the theoretical maximum captured, with deliberately simple forecasts

Annual energy cost of one home (EUR, including battery wear)

Home	Tariff (import / export)	Default settings	Smart EV charger	Our controller	Saving vs default
Lisbon, PV + battery + EV	Three-period / hourly market	340	232	182	**158**
Lisbon, PV + battery + EV	Two-period / hourly market	342	238	195	**147**
Madrid, PV + battery + EV	PVPC / hourly market	317	181	131	**186**
Madrid, PV + battery + EV	Fixed offer / hourly market	276	177	110	**166**
Lisbon, PV + battery	Three-period / hourly market	10	10	-50	**60**
Madrid, PV + battery	PVPC / hourly market	25	25	-71	**96**
Lisbon, PV + battery	Any / fixed 0.01 €/kWh	53–64	53–64	57–66	≈ 0

What it means for an installer or retailer

- **The value is real where exports are paid at market prices** and largest in homes with an EV. Homes on a fixed export price gain little from battery scheduling, so we would not sell it to them.
- **Most installed batteries run on default settings today.** The saving can be passed to customers, sold as a monthly service, or used to win new sales against utility offers.
- **No hardware.** The engine works through the inverter and charger interfaces you already use, and can run read-only first.

Proposed next step

A free, read-only analysis on 50–100 of your installations over a few weeks. We compute what the controller would have done; nothing is switched. You receive a per-home report of the saving before deciding on a pilot.

Method: one home per city simulated hour by hour through 2025: 5 kWp solar (30° south), 10 kWh / 5 kW battery, 3,500 kWh/yr household load, EV of 2,350 kWh/yr with a weekday commute, 3.7 kW charger, 6.9 kVA contract. Prices: OMIE day-ahead 2025 (PT and ES zones). Tariffs: ERSE 2026 regulated BTN prices with VAT (PT); 2026 2.0TD network charges and cargos with taxes (ES). Export at the hourly market price minus €10/MWh, or fixed. Solar: NASA POWER 2025 radiation. Forecasts use only data available at decision time. "Smart EV charger" follows solar surplus and the cheapest known hours. Not modelled: Spain's monthly compensation cap, 15-minute settlement. Results for a real fleet will differ; the read-only analysis exists to measure them.